Sanaka Sai Lolitha

cs24btech11055@iith.ac.in | ssailolitha@gmail.com | +91 9347716055 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.22
Narayana Jr. college	Class XII (Board of Intermediate Education,Andhra Pradesh)	2024	96.00%
Viswabharathi (EM) High School	Class X (Board of Secondary Education,Andhra Pradesh)	2022	97.20%

SKILLS

Languages: C,C++,Python,Java,SQL,Assembly,HTML

Databases: PostgreSQL

Frameworks: OpenCV,NumPy

DevTools: Github

RELEVANT COURSES

Mathematics: Calculus,Probability and Random Variables,Linear Algebra,Transform Techniques,Statistics

Computer Science: Discrete Mathematics,Artificial Intelligence,Software Development Fundamentals,Theory of Computation,Operating Systems I,Operating Systems II,Computer Architecture,Data Structures,Algorithms

Database Courses: Database Management Systems I,Database Management Systems II

PROJECTS

Infinite Precision Arbitrary Arithmetic:

- Implemented Integer and Float types using string digit blocks with custom add,subtract,multiply and divide algorithms
- --->gitlink-https://github.com/minny-creator/arbitaryarithmetic

Clustering(GITHUB)

- Implemented k-mean clustering,Tried to identify k points such that points are distributed as much equally as possible into those k-clusters.
- Strategy:Initially assumed random k-points as k-mean points,and then distributed all points into those k-groups,calculated mean of all points in each group,based on the closeness between mean point and initially assumed point for that group,If all groups satisfy closeness criterion ,then considered those k-points as k-means.

Text search engine(GITHUB)

- It searches given files for a keyword in the files efficiently and returns the files which has given keyword.
- It uses bloomfilters for each text file to store words of each file in encoded manner.
- If bloomfilters returns positive,it uses Rabin-karp algorithm to search each word of file.

Real-Time Auction Website | Github

- Engineered a **real-time auction platform** using react ,Node.js ,Express, and Socket.IO,enabling synchronized live bidding and auction-state updates across concurrency participants.
- Implemented **bid validation ,automated winner determination ,and dynamic auction control,** with PostgreSQL and prisma ORM for persistent team, question ,and auction-state management.
- Built JSON Web Tokens-based admin authentication and a responsive dashboard for managing teams ,questions, live auction states, and analytics.